

## Linear Algebra

**Exercise 1.** The characteristic polynomial of a  $3 \times 3$  matrix  $A$  is  $\lambda^3 + 2\lambda^2 - 15\lambda$ .

- (a) What is  $\text{Tr}(A)$ ? (b) Is  $A$  invertible? (c) Is  $A$  diagonalizable?

**Solution 1.**

- (a) For an  $n \times n$  matrix, the trace is  $-1$  times the coefficient of  $\lambda^{n-1}$  in the characteristic polynomial, so here the trace is 2. (This is if you take the characteristic polynomial to be  $\det(\lambda I - A)$ , so that the characteristic polynomial is always monic. If you take it to be  $\det(A - \lambda I)$ , then the trace is instead multiplied by  $(-1)^{n-1}$ .)
- (b) No,  $A$  is not invertible. The determinant of  $A$  is the constant coefficient of the characteristic polynomial, which is 0. Alternatively, you can see that 0 is a root of the characteristic polynomial, so 0 is an eigenvalue of  $A$ .
- (c) Yes,  $A$  is diagonalizable. The roots of the characteristic polynomial are 0, 3, and -5. Since  $A$  has distinct eigenvalues, it is diagonalizable. (Since the eigenvalues are all rational, it does not matter whether you consider the base field to be  $\mathbb{Q}$ ,  $\mathbb{R}$ , or  $\mathbb{C}$ . If you had irrational eigenvalues, whether  $A$  was diagonalizable might depend on what you consider the base field to be.)

**Exercise 2.** For each of the following, either come up with an example, or prove that no example exists.

- (a) A  $2 \times 2$  matrix over  $\mathbb{R}$  with no real eigenvectors.
- (b) A  $2 \times 2$  matrix over  $\mathbb{R}$  with exactly 1 real eigenvector (up to scaling).
- (c) A  $2 \times 2$  matrix over  $\mathbb{C}$  with no complex eigenvectors.
- (d) A  $2 \times 2$  matrix over  $\mathbb{C}$  with exactly 1 complex eigenvector (up to scaling).

**Solution 2.**

- (a) The classic example is the  $90^\circ$  rotation matrix:

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

- (b) The classic example is the shearing transformation (or transvection):

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

- (c) This is not possible, see problem 4 below.
- (d) The same example from part (b) still works here.

**Exercise 3.** The **Cayley-Hamilton Theorem** says that a matrix  $A$  always satisfies its own characteristic polynomial. What that means is that if you write out the characteristic polynomial  $f(x)$ , and plug in the matrix  $A$  (even though it's a matrix and not a number), you always get the 0 matrix:  $f(A) = 0$ .

- (a) Suppose  $A$  is diagonalizable matrix over  $\mathbb{C}$ . Prove the Cayley-Hamilton Theorem for  $A$ . (Hint: you can prove a matrix is the zero matrix by showing it acts by 0 on a basis.)
- (b) Look at your example from Exercise 2 (b). Verify that it satisfies the Cayley-Hamilton Theorem.
- (c) Find an example of an  $n \times n$  matrix that satisfies a polynomial of degree less than  $n$ .

**Solution 3.**

- (a) Since  $A$  is diagonalizable, we can find a basis of eigenvectors  $v_i$  for  $A$ . For each  $v_i$  we have  $Av_i = \lambda_i v_i$ , and so for any polynomial  $g$ ,  $g(A)v_i = g(\lambda_i)v_i$ . Now, if we choose the characteristic polynomial  $f$ , we get  $f(A)v_i = f(\lambda_i)v_i = 0$  because  $\lambda_i$  is a root of  $f$ . Thus,  $f(A)$  acts by 0 on a basis, so  $f(A)$  is the zero matrix.
- (b) Your exact computation may vary if you chose a different matrix. For the matrix above, the characteristic polynomial is  $f(x) = (x - 1)^2$ . When we plug in, we get

$$f(A) = (A - I)^2 = \left( \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right)^2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}^2 = 0.$$

- (c) The  $n \times n$  identity matrix is one example, since it satisfies the degree 1 polynomial  $x - 1$ .

**Exercise 4. (August 2014 Problem 3, parts (a)-(c))**

- (a) Let  $V$  be a finite-dimensional vector space over  $\mathbb{C}$ , and let  $T: V \rightarrow V$  be a linear transformation. Show that  $T$  has an eigenvector.
- (b) Give an example of a nonzero finite-dimensional vector space  $V$  over  $\mathbb{R}$  and a linear transformation  $T: V \rightarrow V$  such that  $T$  does not have an eigenvector.
- (c) Does a linear transformation of an infinite-dimensional vector space over  $\mathbb{C}$  have to have an eigenvector? Either prove this is the case, or give an example of a linear transformation of an infinite-dimensional vector space with no eigenvector.

## Solution 4.

- (a) Let  $\chi(x)$  be the characteristic polynomial of  $T$ . Then by the fundamental theorem of algebra,  $\chi$  has a root  $\lambda$ . This means that  $\det(T - \lambda \text{Id}) = 0$ , so  $T - \lambda \text{Id}$  has nontrivial kernel. For any  $v \in \ker(T - \lambda \text{Id})$ , we have  $Tv = \lambda v$ , so  $v$  is an eigenvector for  $T$ .
- (b) The classic example is a rotation by  $\frac{\pi}{2}$  radians:

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

- (c) This does not have to be the case. Consider the vector space  $V$  whose elements are sequences of complex numbers. So a vector looks like  $v = (z_1, z_2, \dots)$ . Let  $T$  be the “right shift” operator, so  $Tv = (0, z_1, z_2, \dots)$ . If we try setting up  $Tv = \lambda v$ , then looking at the first entry tells us the only possible value of  $\lambda$  is 0. But, you can check that  $T$  has trivial kernel, so it can’t have any eigenvectors.

## Exercise 5.

- (a) Does every monic degree 2 polynomial over  $\mathbb{C}$  occur as the characteristic polynomial of a  $2 \times 2$  matrix with complex entries?
- (b) Does every monic degree 2 polynomial over  $\mathbb{R}$  occur as the characteristic polynomial of a  $2 \times 2$  matrix with real entries?

## Solution 5.

- (a) Yes. Since every degree 2 polynomial factors over  $\mathbb{C}$ , we can write a general degree 2 polynomial as  $(x - \alpha)(x - \beta)$  for some  $\alpha, \beta \in \mathbb{C}$ . Then we can choose the diagonal matrix with entries  $\alpha, \beta$  and this will have the desired characteristic polynomial.
- (b) Yes, but more subtle. The argument above works for any polynomial that factors over  $\mathbb{R}$ , but not every degree 2 polynomial does factor over  $\mathbb{R}$ . Suppose we have a target polynomial  $x^2 + Bx + C$ , we know (see problem 1) that the characteristic polynomial of

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

is  $x^2 - \text{Tr}(A)x + \det(A) = x^2 - (a+d)x + (ad-bc)$ . Then setting  $-(a+d) = B$ , we can solve  $a = -(B+d)$ . Substituting this into  $ad-bc = C$ , we get  $C = -(B+d)d-bc = -d^2-Bd-bc$ , which we can rearrange to  $d^2 + Bd + C = -bc$ . Now, for any given value of  $d$ , we can find (infinitely many) values of  $b, c$  satisfying this equation, so not only can we find a real matrix with a given characteristic polynomial, but we have a 2-dimensional space of

choices how to do so. A particularly simple choice is given by  $d = -B$  (so  $a = 0$ ),  $c = 1$ ,  $b = -C$ , i.e. the matrix

$$\begin{pmatrix} 0 & -C \\ 1 & -B \end{pmatrix}.$$

This is called the *companion matrix* to the polynomial, and it turns out that every matrix with this given characteristic polynomial is similar over  $\mathbb{R}$  to this matrix.

**Exercise 6. (January 1983 Problem 1)** Suppose  $V$  is a finite dimensional vector space over  $\mathbb{C}$ , and  $T: V \rightarrow V$  is a linear operator. Suppose the subspace of  $V$  spanned by all eigenvectors of  $T$  is 1-dimensional. Prove that the minimal polynomial of  $T$  is  $(x - \alpha)^n$ , where  $n = \dim V$  and  $\alpha$  is some complex number.

**Solution 6. not using Jordan normal form:** the characteristic polynomial of  $T$  is a degree  $n$  polynomial whose only root is  $\alpha$ , so it must be  $(x - \alpha)^n$ . By the Cayley-Hamilton theorem,  $T$  satisfies its characteristic polynomial, so the minimal polynomial of  $T$  divides  $(x - \alpha)^n$ . We just need to check that  $(T - \alpha \text{Id})^k \neq 0$  for any  $k < n$ . Notice that  $\dim \ker(T - \alpha \text{Id}) = 1$ , and for any linear transformations  $A, B$ ,  $\dim \ker(AB) \leq \dim \ker(A) + \dim \ker(B)$ . Thus,  $\dim \ker((T - \alpha \text{Id})^k) \leq k < n$ , so  $(T - \alpha \text{Id})^k \neq 0$ .

**Using Jordan normal form:** if we take the Jordan normal form of  $T$ , all the Jordan blocks have  $\alpha$  down the diagonal. However, if there is more than one block, then  $T$  will have two linearly independent eigenvectors, so actually the Jordan form of  $T$  is

$$\begin{pmatrix} \alpha & 1 & 0 & \dots & 0 \\ 0 & \alpha & 1 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \dots & \dots & \alpha & 1 \\ 0 & \dots & \dots & 0 & \alpha \end{pmatrix}.$$

From here, one can compute that  $(T - \alpha \text{Id})^n = 0$ , but  $(T - \alpha \text{Id})^k \neq 0$  for any  $k < n$ . (Indeed, for any upper triangular  $n \times n$  matrix  $W$  with 0 on the diagonal,  $W^n = 0$ .)

**Exercise 7. (January 2022 Problem 2, parts (a),(b),(d))** For each of the following statements, either prove it or provide a counterexample.

- (a) Every matrix of finite order in  $\text{GL}_n(\mathbb{C})$  is diagonalizable.
- (b) Every matrix of finite order in  $\text{GL}_n(\mathbb{Q})$  is diagonalizable.
- (d) Every matrix of finite order in  $\text{GL}_n(\overline{\mathbb{F}}_p)$  is diagonalizable. (We write  $\overline{\mathbb{F}}_p$  for the algebraic closure of the finite field  $\mathbb{F}_p$ .)

**Solution 7.**

- (a) Suppose  $A \in \text{GL}_n(\mathbb{C})$  has finite order, so that  $A^k = I$  for some  $k$ . Then we know the minimal polynomial of  $A$  divides  $x^k - 1$ . Now, over  $\mathbb{C}$ ,  $x^k - 1$  splits into  $k$  different linear factors, so the minimal polynomial also splits into different linear factors. Since the exponents on the linear factors of the minimal polynomial are the sizes of maximal Jordan blocks, this tells us that the size of a maximal Jordan block is 1, which is the same as saying  $A$  is diagonalizable.
- (b) Everyone's favorite counterexample:

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

- (d) The problem with our proof from part (a) is that we said  $x^k - 1$  splits into *distinct* linear terms over  $\mathbb{C}$ , but that need not be true in  $\overline{\mathbb{F}}_p$ . For example,  $x^p - 1$  definitely doesn't. So, take the example of

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

over  $\overline{\mathbb{F}}_2$ .

**Exercise 8. (January 2017 Problem 2)** Let  $n > 0$  be an integer. Let  $F$  be a field of characteristic zero,  $V$  a vector space over  $F$  of dimension  $n$ , and  $T: V \rightarrow V$  an invertible  $F$ -linear map such that  $T^{-1} = T$ . Denote by  $W$  the vector space of linear transformations from  $V$  to  $V$  that commute with  $T$ . Find a formula for  $\dim(W)$  in terms of  $n$  and the trace of  $T$ .

**Solution 8.** Since  $T^{-1} = T$ , we see that  $T^2 - \text{Id}_n = 0$ , so  $T$  satisfies the polynomial  $x^2 - 1 \in F[x]$ . Therefore, the minimal polynomial of  $T$  must divide  $x^2 - 1 = (x - 1)(x + 1)$ . So the only possible eigenvalues of  $T$  are 1 and  $-1$ . Since the minimal polynomial of  $T$  has distinct roots, we can choose a basis so that  $T$  is a diagonal matrix with only 1s and  $-1$ s on the diagonal. Furthermore, we can rearrange the basis so that  $T$  has the block matrix form

$$T = \begin{pmatrix} \text{Id}_a & 0 \\ 0 & -\text{Id}_b \end{pmatrix}$$

where  $a + b = n$ . So the trace of  $T$  is  $\text{Tr}(T) = a - b$ .

We want to understand matrices that commute with  $T$ . Given an  $n \times n$  matrix  $S$ , we consider it as a block matrix of the form  $S = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$  where  $A$  has size  $a \times a$  and  $D$  has size  $b \times b$ . Then multiplying by  $T$  on either side gives

$$TS = \begin{pmatrix} A & B \\ -C & -D \end{pmatrix} \quad \text{and} \quad ST = \begin{pmatrix} A & -B \\ C & -D \end{pmatrix}.$$

Therefore,  $S \in W$  if and only if  $B = -B = 0$  and  $C = -C = 0$ . Since there are no constraints on  $A$  and  $D$ , we see that  $\dim(W) = a^2 + b^2 = \frac{1}{2}((a+b)^2 + (a-b)^2) = \frac{1}{2}(n^2 + \text{Tr}(T)^2)$ .

**Exercise 9. (January 2025 Problem 4)** Let  $V$  be a dimension 4 vector space over  $\mathbb{C}$  and  $\phi: V \rightarrow V$  an operator. Suppose that a dimension 2 subspace  $W$  of  $V$  is invariant in the sense that  $\phi(W) \subset W$ ; then  $\phi$  induces operators  $W \rightarrow W$  and  $V/W \rightarrow V/W$ .

Suppose that the Jordan form of  $\phi$  on both  $W$  and  $V/W$  is a single  $2 \times 2$  Jordan block with eigenvalue 0. Find all possible Jordan forms of  $\phi$  on  $V$ .

(For full credit, you need to provide examples that show your answers indeed can happen and prove that there are no other possibilities.)

**Solution 9.** We know the only possible eigenvalue of  $\phi$  is 0, and that  $\phi$  must have a Jordan block of size at least 2. There are four possible Jordan forms satisfying those conditions, and we will show that only three can arise. Namely, the Jordan form of  $\phi$  can be: a single  $4 \times 4$  block; a  $1 \times 1$  block and a  $3 \times 3$  block; or two  $2 \times 2$  blocks. That is, the Jordan matrix of  $\phi$  can be any of

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \text{or} \quad \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Here are two helpful things for approaching this problem. First, saying we know how  $\phi$  acts on  $W$  and on  $V/W$  is basically the same as saying we know 3 out of 4 blocks of a block matrix representing  $\phi$ . Specifically, if we choose a basis  $v_1, v_2, v_3, v_4$  so that  $v_1, v_2$  are a Jordan basis for  $W$ , and  $v_3 + W, v_4 + W$  are a Jordan basis for  $V/W$ , then we know the matrix of  $\phi$  acting on  $v_1, v_2$  is a  $2 \times 2$  Jordan block. We also know that  $\phi$  acting on  $v_3, v_4$  is a  $2 \times 2$  Jordan block modulo  $W$ , which is the same as saying it's a  $2 \times 2$  Jordan block plus some linear combination of  $v_1$  and  $v_2$ . This means that the matrix of  $\phi$  with respect to this basis look like

$$\begin{pmatrix} 0 & 1 & * & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

where the  $*$  represent unknown entries. So, our task is essentially to understand all the possible Jordan forms that can come about for different values of the  $*$  entries.

The second helpful thing is what exactly you're looking for to determine the Jordan form of  $\phi$ . To get a matrix in Jordan form, you need a basis of Jordan chains, and a Jordan chain of eigenvalue  $\lambda$  is a sequence of vectors  $v_k, v_{k-1}, \dots, v_1$  such that  $\phi(v_k) = \lambda v_k + v_{k-1}$ . (This is where the staircase shape of Jordan matrices comes from.) The general way to build a Jordan chain is to start with  $v_k$  which you know is a generalized eigenvector of eigenvalue  $\lambda$ , and recursively define  $v_{i-1} = \phi(v_i) - \lambda v_i = (\phi - \lambda I)v_i$ .

So, in this case, to build our Jordan chain we should start with  $v_4$  and recursively work our way up. We know  $\phi(v_4) = v_3 + w$  for some  $w \in W$ . We can change basis by declaring  $v'_3 = v_3 + w$ , so without loss of generality we may assume  $w = 0$ . Now, there are several possibilities for  $\phi(v_3)$ :

- We could have  $\phi(v_3) = 0$ . In this case,  $v_1, v_2, v_3, v_4$  is already a Jordan basis for  $\phi$ , and it has two Jordan blocks of size 2.
- Perhaps  $\phi(v_3) \neq 0$  but  $\phi(\phi(v_3)) = 0$ , that is  $\phi(v_3) \in \ker \phi$ . In this case, we know  $\phi(v_3) = cv_1$  for some nonzero scalar  $c$ . In this case, notice that  $\phi(cv_2 - v_3) = cv_1 - cv_1 = 0$ , so  $v_1$  and  $cv_2 - v_3$  are two linearly independent elements of  $\ker \phi$ . Thus,  $cv_2 - v_3, cv_1, v_3, v_4$  is a Jordan basis for  $\phi$ , having one Jordan block of size 1 and one of size 3.
- Perhaps  $\phi(v_3) \notin \ker \phi$ . In this case, we can change basis by  $v'_2 = \phi(v_3)$ ,  $v'_1 = \phi(v'_2)$ , and the basis  $v'_1, v'_2, v_3, v_4$  is a Jordan basis for  $\phi$ , having one Jordan block of size 4.

It is a simple matter to translate this proof that these are the only possibilities into examples of all three happening. For the Jordan block of size 4, we can take  $W = \text{span}(e_1, e_2)$ , for the Jordan blocks of sizes (1,3) we can take  $W = \text{span}(e_1 + e_3, e_2)$ , and for the two Jordan blocks of size 2 we can again take  $W = \text{span}(e_1, e_2)$ .

(There is a way to approach this question as a question about  $\mathbb{C}[x]$ -modules, where subtly what this question is asking is almost but not quite to compute an Ext group.)